A Learning-Controlled Elephant Random Walk:
Martingale Structure, Diffusive Limits and Reproducible Evidence

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Abstract

The elephant random walk is a non-Markovian stochastic process whose next increment is drawn from its complete history and is then repeated or reversed. This article introduces a learning-controlled elephant random walk (LC-ERW), in which the memory parameter is no longer fixed but is selected online by a predictable learning rule. The construction is intended as a mathematically tractable bridge between long-memory random walks and artificial-intelligence mechanisms for adaptive exploration. The main result is a martingale normalisation valid for arbitrary predictable memory policies. Under a stable learning condition, namely convergence of the learned memory coefficient to a diffusive limit smaller than one half, the LC-ERW satisfies a central limit theorem with the same variance form as an elephant random walk with the limiting memory coefficient. The proof is based on triangular martingale convergence and explicit control of the predictable quadratic variation. Reproducible R experiments are provided to check the variance prediction and to illustrate the transition from adaptive behaviour to the limiting Gaussian regime.

Keywords: elephant random walk; artificial intelligence; online learning; martingales; central limit theorem; non-Markovian random walk.

1 Introduction

Random walks with memory provide simple models in which local moves are affected by a long historical record. The elephant random walk (ERW), introduced by Schütz and Trimper (2004), is a canonical example: at each time the walker selects one of its past increments and either repeats or reverses it. This mechanism generates diffusive, critical and superdiffusive regimes, depending on the memory parameter. Subsequent mathematical work has developed central limit theorems, martingale methods, urn connections and variants with restricted memory, random step sizes and delays (Coletti et al., 2017; Bercu, 2018; Baur and Bertoin, 2016; Gut and Stadtmüller, 2021, 2023; Dedecker et al., 2023; Aguech, 2024b,a).

Artificial intelligence (AI), particularly online learning and reinforcement learning, is also concerned with sequential decisions under uncertainty. In many algorithms, exploration is controlled by a parameter that is updated from the past. Yet the usual random-walk component of such methods is often Markovian or externally scheduled. This paper studies a complementary question: what happens when the memory parameter of an ERW is itself produced by a learning rule?

The contribution is a learning-controlled elephant random walk (LC-ERW). The walker has complete elephant memory, but the probability of repeating a recalled increment is chosen from the past by an arbitrary predictable learning algorithm. This includes deterministic schedules, online estimators, stochastic approximation recursions and bounded policy maps. The model is deliberately minimal: it is rich enough to represent AI-style adaptation, but constrained enough to support rigorous proofs.

The results are as follows. First, for any predictable memory policy, the LC-ERW admits an exact martingale normalisation. Secondly, if the learned memory coefficient converges fast enough to a diffusive value $a < 1/2$, then the normalised position $S_n/\sqrt{n}$ converges to a centred normal distribution with variance $(1-2a)^{-1}$. Thirdly, a corollary gives a direct stability condition for online-learning policies: any convergent policy satisfying a mild summability condition inherits the diffusive ERW limit theorem. The numerical section supplies reproducible R code that checks the predicted variance.

The manuscript is not a claim of peer-reviewed novelty. Rather, it states a precise candidate contribution, proves it under transparent assumptions and provides scripts that make the quantitative claims testable.

2 Related Literature

Schütz and Trimper (2004) introduced the ERW as a solvable non-Markovian model with long-range memory. Limit theorems and related asymptotic properties were developed by Coletti et al. (2017), while Bercu (2018) showed that martingale methods give a particularly effective route to strong laws, quadratic strong laws, laws of the iterated logarithm and non-Gaussian behaviour in the superdiffusive regime.

The connection with urn models was clarified by Baur and Bertoin (2016), and extensions have since become an active topic. Gut and Stadtmüller (2021) and Gut and Stadtmüller (2023) discuss variants with modified memory mechanisms. Dedecker et al. (2023) study random step sizes and rates of convergence, while Aguech (2024b) and Aguech (2024a) examine gradually increasing memory, random step sizes and delays. These papers motivate the present study but do not, to the best of our knowledge, formulate the memory parameter itself as the output of a general predictable learning rule.

Table 1 summarises the distinction. The proposed LC-ERW does not claim to dominate these models; rather, it isolates a different source of adaptivity. The increments remain the classical elephant increments, but the repeat–reverse coefficient is updated from the observed past.

Table 1: Position of the LC-ERW relative to common ERW variants.

Model family	Adapted component	Typical mathematical object
Classical ERW	fixed memory parameter	non-Markovian walk
Restricted-memory ERW	recalled time set	memory window or memory growth
Random-step ERW	jump magnitude	random increments with ERW signs
Delayed ERW	move-or-stay decision	ERW with zeros
Pólya-type urn view	colour replacement dynamics	balanced urn representation
LC-ERW	predictable repeat probability	online policy $p_n = \ell(\theta_n)$

3 The Learning-Controlled Model

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space with filtration $(\mathcal{F}_n)_{n\geq 1}$. The position of the walker after n steps is

$$S_n = \sum_{j=1}^{n} X_j, \qquad X_j \in \{-1, 1\}.$$

The first increment satisfies $\mathbb{P}(X_1 = 1) = q$ and $\mathbb{P}(X_1 = -1) = 1 - q$, where $q \in [0, 1]$.

For $n \geq 1$, let K_n be conditionally uniform on $\{1, \ldots, n\}$ given $\mathcal{F}_n$, and let p_n be an $\mathcal{F}_n$-measurable random variable taking values in $(0, 1)$. Conditional on $\mathcal{F}_n$ and K_n, define

$$X_{n+1} = \begin{cases} X_{K_n}, & \text{with probability } p_n, \\ -X_{K_n}, & \text{with probability } 1 - p_n. \end{cases}$$

The process p_n is the learning-controlled memory policy. In AI language, it may be written as

$$p_n = \ell(\theta_n),$$

where θ_n is the state of an online learning algorithm and $\ell : \mathbb{R}^d \to (0, 1)$ is a bounded policy link, for example a logistic map. Set

$$a_n = 2p_n - 1.$$

The coefficient $a_n \in (-1, 1)$ is the learned memory coefficient. Positive values favour repetition of the recalled step, negative values favour reversal, and $a_n = 0$ corresponds to no directional memory in conditional mean.

Assumption 1 (Stable diffusive learning)**.** *There exist constants $a \in (-1, 1/2)$, $\varepsilon \in (0, 1)$, $\bar{a} < 1/2$ and $D < \infty$ such that, almost surely,*

$$a_n \in [-1 + \varepsilon, 1 - \varepsilon] \quad \text{and} \quad a_n \leq \bar{a} \quad \text{for all } n,$$

and

$$\sum_{n=1}^{\infty} \frac{|a_n - a|}{n} \leq D.$$

Remark 1. The summability condition is satisfied whenever $a_n \to a$ at any polynomial rate, for instance $|a_n - a| = O(n^{-\rho})$ with $\rho > 0$. It is deliberately stated pathwise, because the learning rule may be deterministic, data-driven or generated by a stochastic approximation algorithm.

4 Main Results

Theorem 1 (Exact martingale normalisation)**.** *Define $A_1 = 1$ and, for $n \geq 1$,*

$$A_{n+1} = \prod_{k=1}^{n} \left(1 + \frac{a_k}{k}\right).$$

Then

$$M_n = \frac{S_n}{A_n}, \qquad n \geq 1,$$

is a martingale with respect to $(\mathcal{F}_n)$.

Proof. Since K_n is conditionally uniform on $\{1, \ldots, n\}$,

$$\mathbb{E}[X_{K_n} \mid \mathcal{F}_n] = \frac{1}{n} \sum_{j=1}^{n} X_j = \frac{S_n}{n}.$$

Using the conditional repeat–reverse rule,

$$\mathbb{E}[X_{n+1} \mid \mathcal{F}_n] = (p_n - (1 - p_n)) \frac{S_n}{n} = a_n \frac{S_n}{n}.$$

Therefore

$$\mathbb{E}[S_{n+1} \mid \mathcal{F}_n] = S_n + a_n \frac{S_n}{n} = \left(1 + \frac{a_n}{n}\right) S_n.$$

By definition, $A_{n+1} = (1 + a_n/n)A_n$. Hence

$$\mathbb{E}\left[\frac{S_{n+1}}{A_{n+1}}\middle|\mathcal{F}_n\right] = \frac{S_n}{A_n} = M_n.$$

The increments are bounded after normalisation on every finite horizon, so integrability is immediate. $\square$

Lemma 1 (Asymptotics of the normalising product)**.** *Under Assumption 1, there is an almost surely finite random variable $L \in (0, \infty)$ such that*

$$A_n = Ln^a(1 + o(1)) \qquad \textit{almost surely.}$$

Moreover, there exist deterministic constants $0 < c_A < C_A < \infty$ such that

$$c_A n^a \le A_n \le C_A n^a \qquad \textit{for all } n \ge 1$$

on the event where Assumption 1 holds.

Proof. Write

$$\log A_n - a \log n = \sum_{k=1}^{n-1} \left\{ \log\left(1 + \frac{a_k}{k}\right) - \frac{a}{k} \right\} + a \left(\sum_{k=1}^{n-1} \frac{1}{k} - \log n \right).$$

The last term converges because the harmonic numbers satisfy $\sum_{k=1}^{n-1} k^{-1} - \log n \to \gamma$, where γ is Euler's constant. For the first term, decompose

$$\log\left(1 + \frac{a_k}{k}\right) - \frac{a}{k} = \frac{a_k - a}{k} + r_k, \qquad r_k = \log\left(1 + \frac{a_k}{k}\right) - \frac{a_k}{k}.$$

Because $a_k \in [-1 + \varepsilon, 1 - \varepsilon]$, the Taylor remainder is uniformly bounded: there is $C_\varepsilon < \infty$ such that $|r_k| \le C_\varepsilon k^{-2}$ for all k. Hence $\sum_k r_k$ converges absolutely, while $\sum_k (a_k - a)/k$ converges absolutely by Assumption 1. Thus $\log A_n - a \log n$ converges to a finite real limit, say $\log L$, which proves $A_n/(n^a) \to L \in (0, \infty)$.

The same estimates give deterministic bounds. Indeed,

$$|\log A_n - a \log n| \le D + C_\varepsilon \sum_{k=1}^{\infty} k^{-2} + |a| \sup_{m \ge 1} \left| \sum_{k=1}^{m} \frac{1}{k} - \log(m + 1) \right|.$$

Exponentiating this bound gives the constants c_A and C_A. $\square$

Lemma 2 (Second moment control)**.** *Under Assumption 1, there is a deterministic constant C_2 such that*

$$\sup_{n \ge 1} \frac{\mathbb{E}[S_n^2]}{n} \le C_2.$$

Proof. Let $V_n = \mathbb{E}[S_n^2]$. Since $X_{n+1}^2 = 1$,

$$\mathbb{E}[S_{n+1}^2 \mid \mathcal{F}_n] = S_n^2 + 2S_n \mathbb{E}[X_{n+1} \mid \mathcal{F}_n] + 1 = \left(1 + \frac{2a_n}{n}\right) S_n^2 + 1.$$

Taking expectations and using $a_n \le \bar{a} < 1/2$ gives

$$V_{n+1} \le \left(1 + \frac{2\bar{a}}{n}\right) V_n + 1.$$

Set $\beta = 2\bar{a} < 1$. Iterating the recursion yields

$$V_n \le V_1 \prod_{j=1}^{n-1} \left(1 + \frac{\beta}{j}\right) + \sum_{k=1}^{n-1} \prod_{j=k+1}^{n-1} \left(1 + \frac{\beta}{j}\right).$$

There is $C_\beta < \infty$ such that $\prod_{j=k+1}^{n-1}(1+\beta/j) \le C_\beta (n/k)^\beta$. Therefore

$$V_n \le C_\beta n^\beta + C_\beta n^\beta \sum_{k=1}^{n-1} k^{-\beta} \le C_2 n,$$

because $\beta < 1$. This proves the claim. $\square$

Lemma 3 (Predictable quadratic variation)**.** *Under Assumption 1,*

$$\frac{A_n^2}{n} \langle M \rangle_n \xrightarrow{\mathbb{P}} \frac{1}{1-2a}.$$

Proof. The martingale increments are

$$\Delta M_{k+1} = \frac{X_{k+1} - a_k S_k / k}{A_{k+1}}.$$

Consequently,

$$\langle M \rangle_n = \sum_{k=1}^{n-1} \frac{\mathbb{E}[(X_{k+1} - a_k S_k/k)^2 \mid \mathcal{F}_k]}{A_{k+1}^2} = \sum_{k=1}^{n-1} \frac{1 - a_k^2 S_k^2 / k^2}{A_{k+1}^2}.$$

We first evaluate the leading term. Put $B_n = A_n / n^a$. By Lemma 1, $B_n \to L \in (0, \infty)$ almost surely. Hence

$$\frac{A_n^2}{n} \sum_{k=1}^{n-1} \frac{1}{A_{k+1}^2} = B_n^2 n^{2a-1} \sum_{k=1}^{n-1} \frac{1}{B_{k+1}^2 (k+1)^{2a}}.$$

Since $B_{k+1}^{-2} \to L^{-2}$ and $2a < 1$, Cesaro summation for regularly varying sequences gives

$$n^{2a-1} \sum_{k=1}^{n-1} \frac{1}{B_{k+1}^2 (k+1)^{2a}} \longrightarrow \frac{L^{-2}}{1-2a}.$$

Multiplication by $B_n^2 \to L^2$ gives

$$\frac{A_n^2}{n} \sum_{k=1}^{n-1} \frac{1}{A_{k+1}^2} \longrightarrow \frac{1}{1-2a} \qquad \text{almost surely.}$$

It remains to show that the correction term is negligible. By Lemma 1, $|a_k| \le 1$ and Lemma 2,

$$\begin{aligned}\mathbb{E}\left[\frac{A_n^2}{n} \sum_{k=1}^{n-1} \frac{a_k^2 S_k^2 / k^2}{A_{k+1}^2}\right] &\le C n^{2a-1} \sum_{k=1}^{n-1} k^{-2a-2} \mathbb{E}[S_k^2] \\ &\le C n^{2a-1} \sum_{k=1}^{n-1} k^{-2a-1}.\end{aligned}$$

The final expression tends to zero in all cases $a < 1/2$: if $a > 0$, the sum is bounded; if $a = 0$, it is $O(\log n)$; and if $a < 0$, it is $O(n^{-2a})$, giving $O(n^{-1})$. Thus the correction term converges to zero in L^1, and hence in probability. The lemma follows. $\square$

Lemma 4 (Lindeberg condition). *Under Assumption 1, for every $\delta > 0$,*

$$\frac{1}{\langle M \rangle_n} \sum_{k=1}^{n-1} \mathbb{E}\left[(\Delta M_{k+1})^2 \mathbf{1}_{\{|\Delta M_{k+1}| > \delta \langle M \rangle_n^{1/2}\}} \middle| \mathcal{F}_k \right] \xrightarrow{\mathbb{P}} 0.$$

Proof. Because $|X_{k+1}| \le 1$, $|a_k| \le 1$, and $|S_k| \le k$,

$$|\Delta M_{k+1}| = \frac{|X_{k+1} - a_k S_k / k|}{A_{k+1}} \le \frac{2}{A_{k+1}}.$$

Lemma 1 gives $A_k \asymp k^a$. If $a \ge 0$, then $\max_{k \le n} |\Delta M_k| = O(1)$; if $a < 0$, then $\max_{k \le n} |\Delta M_k| = O(n^{-a})$. On the other hand, Lemma 3 implies

$$\langle M \rangle_n^{1/2} \asymp n^{1/2 - a} \qquad \text{in probability.}$$

Therefore

$$\frac{\max_{k \le n} |\Delta M_k|}{\langle M \rangle_n^{1/2}} \xrightarrow{\mathbb{P}} 0.$$

With probability tending to one, every indicator in the Lindeberg sum is then zero, which proves the assertion. □

Theorem 2 (Diffusive central limit theorem). *Under Assumption 1,*

$$\frac{S_n}{\sqrt{n}} \xrightarrow{\mathcal{D}} \mathcal{N}\left(0, \frac{1}{1 - 2a}\right).$$

Proof. The martingale $M_n = S_n / A_n$ has increments satisfying the conditional Lindeberg condition by Lemma 4. Since $\langle M \rangle_n \to \infty$ in probability by Lemmas 1 and 3, the martingale central limit theorem (Hall and Heyde, 1980, Theorem 3.2, for instance) gives

$$\frac{M_n - M_1}{\langle M \rangle_n^{1/2}} \xrightarrow{\mathcal{D}} \mathcal{N}(0, 1).$$

Moreover $M_1 / \langle M \rangle_n^{1/2} \to 0$ in probability. Hence

$$\frac{M_n}{\langle M \rangle_n^{1/2}} \xrightarrow{\mathcal{D}} \mathcal{N}(0, 1).$$

Lemma 3 states that

$$\frac{A_n^2}{n} \langle M \rangle_n \xrightarrow{\mathbb{P}} \frac{1}{1 - 2a}.$$

Since $S_n = A_n M_n$, Slutsky's theorem yields

$$\frac{S_n}{\sqrt{n}} = \frac{M_n}{\langle M \rangle_n^{1/2}} \left(\frac{A_n^2 \langle M \rangle_n}{n} \right)^{1/2} \xrightarrow{\mathcal{D}} \mathcal{N}\left(0, \frac{1}{1 - 2a}\right).$$

□

Corollary 1 (Stability of convergent learning policies). *Let $\theta_n \in \mathbb{R}^d$ be an $\mathcal{F}_n$-predictable learning state and let $\ell : \mathbb{R}^d \to (\varepsilon, 1 - \varepsilon)$ be Lipschitz. Put*

$$p_n = \ell(\theta_n), \qquad a_n = 2p_n - 1.$$

If $\theta_n \to \theta_\star$ almost surely and there is $D_\theta < \infty$ such that

$$\sum_{n=1}^{\infty} \frac{\|\theta_n - \theta_\star\|}{n} \le D_\theta \qquad \textit{almost surely,}$$

with $a = 2\ell(\theta_\star) - 1 < 1/2$, and if there exists $\bar{a} < 1/2$ such that $2\ell(\theta_n) - 1 \le \bar{a}$ for all n, then the LC-ERW satisfies the central limit theorem in Theorem 2.

Proof. By Lipschitz continuity,

$$|a_n - a| \le 2\,\mathrm{Lip}(\ell)\|\theta_n - \theta_\star\|.$$

The displayed summability condition implies Assumption 1, and Theorem 2 applies. □

Proposition 1 (A damped empirical-drift learner)**.** *Fix $\theta_\star \in \mathbb{R}$, $c, \beta \in \mathbb{R}$, $\rho > 0$, $\delta > 0$, and let $\ell : \mathbb{R} \to (\varepsilon, 1-\varepsilon)$ be Lipschitz. Define the predictable learning state*

$$\theta_n = \theta_\star + c(n+1)^{-\rho} + \beta \frac{S_n}{n^{1+\delta}}, \qquad n \ge 1,$$

and set $p_n = \ell(\theta_n)$. Suppose that $a = 2\ell(\theta_\star) - 1 < 1/2$ and that $2\ell(\theta_n) - 1 \le \bar{a} < 1/2$ for all n. Then the resulting LC-ERW satisfies

$$\frac{S_n}{\sqrt{n}} \xrightarrow{\mathcal{D}} \mathcal{N}\left(0, \frac{1}{1-2a}\right).$$

Proof. The rule is predictable because S_n is $\mathcal{F}_n$-measurable. Since $|S_n| \le n$,

$$|\theta_n - \theta_\star| \le |c|(n+1)^{-\rho} + |\beta| n^{-\delta}.$$

Therefore

$$\sum_{n=1}^{\infty} \frac{|\theta_n - \theta_\star|}{n} \le |c| \sum_{n=1}^{\infty} \frac{1}{n(n+1)^\rho} + |\beta| \sum_{n=1}^{\infty} \frac{1}{n^{1+\delta}} < \infty.$$

The result follows from Corollary 1. This proposition gives a concrete learning rule whose correction term uses the observed empirical drift but is damped strongly enough that the learned memory coefficient remains in the stable diffusive class. □

5 Examples of Learning Rules

Deterministic annealing. Let $p_n = p_\star + c(n+1)^{-\rho}$, truncated to $(\varepsilon, 1-\varepsilon)$, where $p_\star < 3/4$ and $\rho > 0$. Then $a_n \to a = 2p_\star - 1 < 1/2$ and the summability condition holds.

Online logistic policy. Let $p_n = \ell(\theta_n)$, where $\ell(x) = \varepsilon + (1-2\varepsilon)/(1+\exp(-x))$, and suppose θ_n is an online estimator converging to $\theta_\star$ at a polynomial rate. The corollary gives the limiting law whenever $\ell(\theta_\star) < 3/4$.

Damped empirical-drift learning. The rule in Proposition 1 is deliberately simple: the learner uses the current empirical drift S_n/n, but the term is multiplied by $n^{-\delta}$. This keeps the policy genuinely data-dependent while making the stability condition verifiable without an additional stochastic approximation theorem.

Stochastic approximation policies. More elaborate controllers can be analysed by first proving convergence of θ_n through standard stochastic approximation arguments (Kushner and Yin, 2003), and then applying Corollary 1. This two-step route is useful when the policy parameter is updated by a projected Robbins–Monro recursion or by a bounded online estimator.

6 Numerical Experiments

The accompanying R scripts simulate the LC-ERW for several policies:

1. a fixed-memory benchmark;
2. a deterministic annealing policy;
3. a logistic learning policy with polynomial convergence;
4. the damped empirical-drift learner in Proposition 1.

For each policy, the scripts estimate the empirical variance of $S_n/\sqrt{n}$, compare it with $(1-2a)^{-1}$, and report Monte Carlo standard errors, confidence intervals, skewness, excess kurtosis and Kolmogorov–Smirnov diagnostics against the predicted Gaussian law. The scripts also generate QQ-plot PDFs for visual inspection. These computations do not prove the theorem; they check whether finite sample behaviour is consistent with the asymptotic prediction and how quickly the variance ratio approaches one as n grows.

7 Discussion

The LC-ERW shows that an AI-style adaptive memory policy can be incorporated into an ERW without destroying its martingale backbone. The main theorem says that, in the diffusive regime, stable learning is asymptotically equivalent to freezing the memory parameter at its learned limit. This is useful in two directions. For probability theory, it extends ERW analysis from fixed or externally prescribed parameters to predictable data-driven policies. For AI, it supplies a rare case in which an adaptive exploration mechanism with complete memory has a transparent asymptotic law.

Several limitations remain. The theorem does not cover critical learning limits $a = 1/2$, superdiffusive limits $a > 1/2$, multidimensional ERW, or learning rules whose parameter does not converge. These cases are likely to require different normalisations and may lead to non-Gaussian limits. Another open direction is statistical inference: estimating the learned memory trajectory p_n from observed LC-ERW paths.

8 Conclusion

This paper formulated a learning-controlled elephant random walk and proved that predictable adaptive memory policies admit an exact martingale normalisation. Under a stable diffusive learning condition, the process obeys a central limit theorem with variance determined by the learned limiting memory coefficient. The result gives a rigorous and reproducible starting point for connecting elephant random walks with AI-controlled stochastic exploration.

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